

Singularity

➤ **Define singularity with examples.**

Ans: A point $z = a$ is called a singular point or a singularity of $f(z)$ if $f(z)$ fails to be analytic at $z = a$.

Example 1:

$f(z) = \frac{1}{(z-1)^2}$ has a singularity at $z = 1$ because

$$f(1) = \frac{1}{(1-1)^2} = \infty$$

i.e., $f(z)$ fails to be analytic at $z = 1$.

Example 2:

$f(z) = \frac{\sin z}{z}$ has a singularity at $z = 0$.

Non-Examples of Singularity

1. $f(z) = z^2 + 3z + 5$

- This function is a polynomial, which is analytic everywhere in the complex plane. Since it has no points where it becomes undefined or fails to be analytic, it **does not** have any singularities.

2. $f(z) = e^z$

- The exponential function is entire, meaning it is analytic everywhere in the complex plane. There is no point where it fails to be defined or analytic, so it **does not** have any singularities.

Types of Singularity:

Isolated Singularity

A point $z = a$ is called an **isolated singularity** of $f(z)$ if $f(z)$ is analytic at each point in the neighborhood of $z = a$ except at $z = a$.

Examples:

1. $f(z) = \frac{1}{z-1}$

- This function has an **isolated singularity** at $z = 1$ because $f(z)$ is analytic everywhere except at $z = 1$.

2. $f(z) = \frac{z+5}{z^2(z^2+9)}$

- This function has **isolated singularities** at $z = 0$ and $z = \pm 3i$.

Non-Isolated Singularity

A function $f(z)$ has a **non-isolated singularity** at $z = a$ if $f(z)$ is non-analytic at **other points** in the neighborhood of $z = a$.

Examples:

1. $f(z) = \frac{1}{\sin(1/z)}$

- This function has $z = 0$ as a **non-isolated singularity** because in the neighborhood of $z = 0$, there exist an **infinite number of points** $z = \frac{1}{k}$ for $k = 106, 109, 112, \dots$ at which $f(z)$ is non-analytic and are very close to $z = 0$.

2. $f(z) = \log z$

- This function has $z = 0$ as a **non-isolated singularity** because every neighborhood of $z = 0$ contains an **infinite number of points**, especially on the **negative part of the real axis**, where $\log z$ is non-analytic.

Isolated Singularity can further be classified as:

i) Pole

ii) Removable Singularity

iii) Essential Singularity

Pole:

A singularity of finite order is called a pole.

Examples:

i) $f(z) = \frac{1}{z-3}$ has a pole of order 1 at $z = 3$.

$$\begin{aligned} \text{ii) } f(z) &= \frac{\sinh z}{z^7} = \frac{1}{z^7} \left\{ z + \frac{z^3}{3!} + \frac{z^5}{5!} + \frac{z^7}{7!} + \dots \right\} \\ &= \frac{1}{z^6} + \frac{1}{3!z^4} + \frac{1}{5!z^2} + \dots \end{aligned}$$

$\Rightarrow f(z)$ has a pole of order 6 at $z = 0$.

iii) $f(z) = \frac{z}{2} + \frac{4}{z^7} + \frac{1}{z^7}$ has a pole of order 7 at $z = 0$.

iv) $f(z) = 2 + \frac{5}{z} + \frac{6}{z^2}$ has a pole of order 2 at $z = 0$.

Essential Singularity:

A singularity that cannot be removed is called an **essential singularity**.

Examples:

i) $f(z) = \frac{1}{z}$ has an essential singularity at $z = 0$.

ii) $f(z) = e^{1/z} = 1 + \frac{1}{z} + \frac{1}{2!z^2} + \frac{1}{3!z^3} + \dots$ has an essential singularity at $z = 0$.

Removable Singularity:

A singularity $z = a$ is called a **removable singularity** for $f(z)$ if, on simplification or as a limiting case, it disappears.

Example:

$f(z) = \frac{\sin z}{z}$ has a singularity at $z = 0$.

But,

$$f(z) = \frac{1}{z} \left\{ z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \dots \right\} = 1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \frac{z^6}{7!} + \dots$$

$$\lim_{z \rightarrow 0} f(z) = 1$$

So, $z = 0$ is a **removable singularity** for $f(z) = \frac{\sin z}{z}$.

Singularity at Infinity: The type of singularity of $f(z)$ at $z = \infty$ is the same as that of $f(\frac{1}{w})$ at $w = 0$.

Examples-1:

The function $f(z) = z^3$ has a pole of order 3 at $z = \infty$, since $f(1/w) = 1/w^3$ has a pole of order 3 at $w = 0$.

Example 2: $f(z) = e^z$

- Consider the function $f(z) = e^z$.
- Using the transformation $w = \frac{1}{z}$, we rewrite the function as:

$$f(1/w) = e^{1/w}.$$

- The function $e^{1/w}$ has an **essential singularity** at $w = 0$ because it exhibits highly irregular behavior in its Laurent series expansion.
- This means $f(z) = e^z$ has an **essential singularity** at $z = \infty$.

Example 3

Function: $f(z) = z^5$

- Using the transformation $w = \frac{1}{z}$, we express the function as:

$$f(1/w) = \left(\frac{1}{w}\right)^5 = \frac{1}{w^5}.$$

- The function $\frac{1}{w^5}$ has a **pole of order 5** at $w = 0$.
- This implies that $f(z) = z^5$ has a **pole of order 5** at $z = \infty$.

Working Rules:

Pole: If there exists a positive integer n such that

$$\lim_{z \rightarrow z_0} (z - z_0)^n f(z) = A \neq 0,$$

then $z = z_0$ is called a **pole of order n** .

- If $n = 1$, then z_0 is called a **simple pole**.
- If $n = 2$, then z_0 is called a **double pole**, and so on.

Example: Let

$$f(z) = \frac{1}{(z-1)^2}$$

then

$$\begin{aligned} & \lim_{z \rightarrow 1} (z-1)^2 f(z) \\ &= \lim_{z \rightarrow 1} (z-1)^2 \cdot \frac{1}{(z-1)^2} = 1 \neq 0. \end{aligned}$$

Thus, $z = 1$ is a **pole of order 2**.

Removable Singularity:

If

$$\lim_{z \rightarrow z_0} f(z)$$

exists, then z_0 is called a **removable singularity** of $f(z)$.

Example:

Let

$$f(z) = \frac{\sin z}{z},$$

then

$$\lim_{z \rightarrow 0} \frac{\sin z}{z} = 1.$$

So, the singular point $z = 0$ is a **removable singularity** of $f(z)$.

Entire / Integral Function:

A function which has no finite singularity and is analytic everywhere except at infinity is called an **Entire** or **Integral function**.

Example:

$$f(z) = e^z, \quad g(z) = \sin z, \quad h(z) = \cosh z$$

are entire functions.

Meromorphic Function:

A function which has **poles as its only singularities** in the finite part of the plane is called a **Meromorphic function**.

Example:

$$f(z) = \frac{z^2 + 1}{z(z + 1)(z^2 + 4)}$$

has singularities at $z = 0, -1, \pm 2i$ as **poles**.

So, it is **meromorphic**, as it is analytic except at $z = 0, -1, \pm 2i$.

Problem:

Discuss the nature of **singularities** of the following functions:

$$f(z) = \frac{e^z}{(z - 1)^3}$$

Solution:

When $(z - 1)^3 = 0$, i.e., $z = 1$, then $f(z) = \infty$.

$\therefore z = 1$ is a **singular point** of $f(z)$.

Now,

$$\begin{aligned}
& \lim_{z \rightarrow 1} (z-1)^3 f(z) \\
&= \lim_{z \rightarrow 1} (z-1)^3 \cdot \frac{e^z}{(z-1)^3} \\
&= \lim_{z \rightarrow 1} e^z \\
&= e^1 = e
\end{aligned}$$

which is **finite**.

$\therefore z = 1$ is a **pole of order 3**.

(ii) Function:

$$f(z) = \frac{z - \sin z}{z^3}$$

Solution:

When $z^3 = 0$, i.e., $z = 0$, then $f(z) = \infty$.

$\therefore z = 0$ is a **singular point** of $f(z)$.

Now,

$$\lim_{z \rightarrow 0} f(z)$$

Since this is of the form $\frac{0}{0}$, we apply **L'Hôpital's Rule**:

$$\lim_{z \rightarrow 0} \frac{z - \sin z}{z^3}$$

Applying L'Hôpital's Rule once:

$$\lim_{z \rightarrow 0} \frac{1 - \cos z}{3z^2}$$

Again applying L'Hôpital's Rule:

$$\begin{aligned} \lim_{z \rightarrow 0} \frac{0 + \sin z}{6z} \\ &= \lim_{z \rightarrow 0} \frac{\sin z}{6z} \\ &= \frac{\cos 0}{6} = \frac{1}{6} \end{aligned}$$

which is **finite**.

$\therefore z = 0$ is a **removable singularity** of $f(z)$.

(iii) Function:

$$f(z) = e^{1/z}$$

Solution:

$$\begin{aligned} f(z) = e^{1/z} &= 1 + \frac{1}{2z} + \frac{1}{2!(2z)^2} + \frac{1}{3!(2z)^3} + \dots \\ &= 1 + \frac{1}{2z} + \frac{1}{8z^3} + \frac{1}{48z^5} + \dots \end{aligned}$$

Clearly, $z = 0$ is an **essential singularity** for $f(z) = e^{1/z}$.

Problem:

For the function $f(z) = \frac{z^8 + z + 2}{(z-1)^3(3z+2)^2}$, locate and name all the singularities in the finite z -plane and determine where $f(z)$ is analytic.

Step 1: Find the Singularities

The singularities occur where the denominator is zero. Solve:

$$(z - 1)^3(3z + 2)^2 = 0$$

This gives:

$$(z - 1)^3 = 0 \quad \text{or} \quad (3z + 2)^2 = 0$$

$$z - 1 = 0 \quad \Rightarrow \quad z = 1$$

$$3z + 2 = 0 \quad \Rightarrow \quad z = -\frac{2}{3}$$

Thus, the singularities in the finite z -plane are $z = 1$ and $z = -\frac{2}{3}$.

Step 2: Determine the nature of the Singularities

Here

$$\lim_{z \rightarrow 1} (z - 1)^3 f(z) = \lim_{z \rightarrow 1} (z - 1)^3 \cdot \frac{z^8 + z + 2}{(z - 1)^3(3z + 2)^2}$$

$$= \lim_{z \rightarrow 1} \frac{z^8 + z + 2}{(3z + 2)^2}$$

$$= \lim_{z \rightarrow 1} \frac{z^8 + z + 2}{(3z + 2)^2} = \frac{1 + 1 + 2}{(3 + 2)^2} = \frac{4}{25} \neq 0$$

$$\lim_{z \rightarrow -\frac{2}{3}} \left(z + \frac{2}{3}\right)^2 f(z) = \lim_{z \rightarrow -\frac{2}{3}} \frac{(3z + 2)^2}{9} \cdot \frac{z^8 + z^4 + 1}{(z - 1)^3(3z + 2)^2}$$

$$= \lim_{z \rightarrow -\frac{2}{3}} \frac{z^8 + z^4 + 1}{9(z - 1)^3}$$

$$= \frac{\left(\frac{2}{3}\right)^8 + \left(\frac{2}{3}\right)^4 + 1}{9\left(-\frac{2}{3} - 1\right)^3} \neq 0$$

$$\begin{aligned}
\lim_{w \rightarrow 0} w^{-3} f\left(\frac{1}{w}\right) &= \lim_{w \rightarrow 0} w^3 \cdot \frac{1 + w^4 + 2w^8}{w^3(1-w)^3(3+2w)^2} \\
&= \lim_{w \rightarrow 0} \frac{1 + w^4 + 2w^8}{(1-w)^3(3+2w)^2} \\
&= \frac{1 + 0 + 0}{(1-0)^3(3+0)^2} \\
&= \frac{1}{1 \times 9} = \frac{1}{9} \neq 0
\end{aligned}$$

$\therefore z = 1$ is a pole of order 3

$z = -\frac{2}{3}$ is a pole of order 2

and $z = \infty$, is a pole of order 3.

Exercise Set: Finding Singularities

Problem 1:

Find and classify all singularities of the function

$$f(z) = \frac{z^3 + 2z}{(z-2)(z^2+1)}$$

Determine where $f(z)$ is analytic.

Problem 2:

Find the singularities of the function

$$f(z) = \frac{e^z}{(z-3)^2(z^2+4)}$$

Classify each singularity as a pole (with order) or an essential singularity.

Problem 3:

Determine the singularities of

$$f(z) = \frac{z^2 + 1}{(z + 1)(z - i)^2}$$

and state where the function is analytic.

Problem 4:

Find the singularities and their types for the function

$$f(z) = \frac{\sin z}{z^2(z^2 - 9)}$$

Identify all the points where the function is analytic.

Problem 5:

Locate and name the singularities of the function

$$f(z) = \frac{(z - 1)^3}{z^4 + 4z^2}$$

State where $f(z)$ is analytic.

Residue and Cauchy's Residue Theorem

Residue at a Pole:

Let $z = a$ be a pole of order m for $f(z)$ then $f(z)$ can be expanded as Laurent's Series as:

$$f(z) = \sum_{n=0}^{\infty} a_n(z-a)^n + \frac{b_1}{z-a} + \frac{b_2}{(z-a)^2} + \dots + \frac{b_m}{(z-a)^m}$$

where $b_m \neq 0$ then the Coeff. of $\frac{1}{z-a}$ ($= b_1$) is called residue of $f(z)$ at $z = a$ which may or may not be zero.

If $z = a$ is a simple pole then residue at $z = a$ is

$$b_1 = \lim_{z \rightarrow a} (z-a)f(z)$$

It is also denoted by $R(f, a) = b_1 = \lim_{z \rightarrow a} (z-a)f(z)$.

If $z = a$ is a pole of order "m" then residue at $z = a$ is

$$R(f, a) = \frac{1}{(m-1)!} \cdot \lim_{z \rightarrow a} \frac{d^{m-1}}{dz^{m-1}} \{(z-a)^m f(z)\}$$

Question: Find the residue of $f(z) = \frac{1}{(z^2+1)^3}$ at $z = i$.

Solution: Clearly $f(z)$ has poles of order 3 at $z = i, -i$

$$\begin{aligned} R(f, i) &= \frac{1}{2!} \cdot \lim_{z \rightarrow i} \frac{d^2}{dz^2} \{ (z - i)^3 f(z) \} \\ &= \frac{1}{2} \cdot \lim_{z \rightarrow i} \frac{d^2}{dz^2} \left\{ (z - i)^3 \cdot \frac{1}{(z + i)^3 (z - i)^3} \right\} \\ &= \frac{1}{2} \cdot \lim_{z \rightarrow i} \frac{d^2}{dz^2} \left\{ \frac{1}{(z + i)^3} \right\} \\ &= \frac{1}{2} \cdot \lim_{z \rightarrow i} \{ (-3)(-4)(z + i)^{-5} \} \\ &= \frac{1}{2} \cdot \frac{12}{(i + i)^5} = \frac{6}{(2i)^5} = \frac{6}{32i^5} \\ &= \frac{6}{32i} = -\frac{6i}{32} = \frac{-3i}{16} \quad \text{Ans.} \\ &\because i^5 = i \quad \text{and} \quad \frac{1}{i} = -i \end{aligned}$$

Question: Find the residues of $f(z) = \frac{2z-5}{(z-1)^2(z+2)}$

Solution: $f(z)$ has poles of order 2 and 1 at $z = 1, -2$ respectively.

$$\begin{aligned} R(f, 1) &= \lim_{z \rightarrow 1} \frac{1}{1!} \cdot \frac{d}{dz} \{(z-1)^2 f(z)\} = \lim_{z \rightarrow 1} \frac{d}{dz} \left\{ \frac{2z-5}{z+2} \right\} \\ &= \lim_{z \rightarrow 1} \frac{(z+2) \cdot 2 - (2z-5) \cdot 1}{(z+2)^2} = \frac{2(1+2) - (2(1) - 5)}{(1+2)^2} \\ &= \frac{6 - 2 + 5}{9} = \frac{9}{9} = 1 \end{aligned}$$

Now

$$\begin{aligned} R(f, -2) &= \lim_{z \rightarrow -2} \{(z+2)f(z)\} \\ &= \lim_{z \rightarrow -2} \left\{ \frac{(z+2)(2z-5)}{(z-1)^2(z+2)} \right\} = \frac{2(-2) - 5}{(-2-1)^2} = \frac{-4-5}{(-3)^2} = \frac{-9}{9} = -1 \end{aligned}$$

Hence $R(f, 1) = 1$, $R(f, -2) = -1$

Problem: Calculate the residue of the function

$$\frac{z^2}{z^2 + a^2} \text{ (take only the positive value of } z\text{),}$$

Soln: Given that $f(z) = \frac{z^2}{z^2 + a^2}$

The poles of $f(z)$ are obtained by solving the equn

$$z^2 + a^2 = 0 \Rightarrow z = \pm ai.$$

Here $z = ai$ is a simple pole.

Residue at $z = ai$ is

$$\begin{aligned} & \lim_{z \rightarrow ai} \left\{ (z - ai) \frac{z^2}{z^2 + a^2} \right\} \\ &= \lim_{z \rightarrow ai} \frac{z^2(z - ai)}{(z + ai)(z - ai)} \\ &= \lim_{z \rightarrow ai} \frac{z^2}{z + ai} \\ &= \frac{-a^2}{2ai} \\ &= \frac{-ai}{2} \end{aligned}$$

Problem:

Find the residues of

$$f(z) = \frac{z^2 - 2z}{(z + 1)^2(z + 4)}$$

at all its poles in a finite plane.

Soln: Given that

$$f(z) = \frac{z^2 - 2z}{(z+1)^2(z+4)}$$

The poles are obtained by solving the equun

$$(z+1)^2(z+4) = 0$$

$$\therefore z = -1, -1, -4$$

Here $z = -1$ is a double pole and $z = -4$ is a simple pole.

Residue at $z = -1$ is

$$\begin{aligned} & \lim_{z \rightarrow -1} \frac{1}{1!} \frac{d}{dz} \{ (z+1)^2 f(z) \} \\ &= \lim_{z \rightarrow -1} \frac{d}{dz} \left\{ \frac{(z+1)^2(z^2 - 2z)}{(z+1)^2(z+4)} \right\} \\ &= \lim_{z \rightarrow -1} \frac{d}{dz} \left(\frac{z^2 - 2z}{z+4} \right) \\ &= \lim_{z \rightarrow -1} \frac{(z+4)(2z-2) - (z^2-2z)(1)}{(z+4)^2} \\ &= \lim_{z \rightarrow -1} \frac{2z^2 + 8z - 2z - 8 - z^2 + 2z}{(z+4)^2} \\ &= \frac{2 - 8 - 8}{(1+4)^2} = \frac{-14}{25} \end{aligned}$$

Residue at $z = -4$ is

$$\begin{aligned} & \lim_{z \rightarrow -4} (z + 4)f(z) \\ &= \lim_{z \rightarrow -4} \left\{ (z + 4) \frac{z^2 - 22}{(z + 1)^2(z + 4)} \right\} \\ &= \lim_{z \rightarrow -4} \frac{z^2 - 22}{(z + 1)^2} \\ &= \frac{16 + 8}{(4 + 1)^2} \\ &= \frac{24}{9} \end{aligned}$$

Exercise Set: Residue at a Pole

Q1. Find the residue of

$$f(z) = \frac{3z + 1}{(z - 2)^2(z + 1)}$$

at $z = 2$ and $z = -1$.

Q2. Find the residue of

$$f(z) = \frac{1}{(z^2 + 4)^2}$$

at $z = 2i$.

Q3. Find the residue of

$$f(z) = \frac{z^2 + 1}{(z - 1)^3}$$

at $z = 1$.

Q4. Find the residue of

$$f(z) = \frac{e^z}{(z + \pi)^2}$$

at $z = -\pi$.

Q5. Find the residue of

$$f(z) = \frac{\sin z}{(z - \frac{\pi}{2})^2}$$

at $z = \frac{\pi}{2}$.

1. Find the residue of $f(z) = \frac{z+1}{z^3(z^2+1)}$ at all its pole in a finite plane.

Solution : Given that $f(z) = \frac{z+1}{z^3(z^2+1)}$

The poles are obtained by solving the equation, $z^3(z^2+1) = 0$

$$\Rightarrow z = 0, \pm i.$$

$\therefore z = 0$ is a pole of order 3 and $z = \pm i$ are pole of order 1.

Residue at $z = 0$, $\lim_{z \rightarrow 0} \frac{1}{2!} \frac{d^2}{dz^2} \{z^3 f(z)\}$

$$= \lim_{z \rightarrow 0} \frac{1}{2} \cdot \frac{d^2}{dz^2} \left\{ z^3 \cdot \frac{z+1}{z^3(z^2+1)} \right\}$$

$$= \lim_{z \rightarrow 0} \frac{1}{2} \frac{d}{dz} \left\{ \frac{(z^2+1) \cdot 1 - (z+1) \cdot 2z}{(z^2+1)^2} \right\}$$

$$= \lim_{z \rightarrow 0} \frac{1}{2} \cdot \frac{d}{dz} \left\{ \frac{z^2+1-2z^2-2z}{(z^2+1)^2} \right\}$$

$$= \lim_{z \rightarrow 0} \frac{1}{2} \frac{d}{dz} \left\{ \frac{1-2z-z^2}{(z^2+1)^2} \right\}$$

$$= \lim_{z \rightarrow 0} \frac{1}{2} \frac{(z^2+1)^2 \cdot (-2-2z) - (1-2z-z^2) \cdot 2(z^2+1) \cdot 2z}{(z^2+1)^4}$$

$$= \frac{1}{2} \frac{1(-2) - 0}{1}$$

$$= \frac{1}{2} (-2)$$

$$= -1$$

Residue at $z = i$, $\lim_{z \rightarrow i} (z-i) f(z)$

$$= \lim_{z \rightarrow i} (z-i) \frac{(z+1)}{z^3(z+i)(z-i)}$$

$$= \frac{1+i}{i^3 \cdot 2i}$$

$$= \frac{1+i}{2}$$

Similarly, residue at $z = -i$ is

$$\frac{1-i}{2} \text{ [Replacing } i \text{ by } -i \text{ in the residue at } z = i]$$

2. Find the residues of the function

$$f(z) = \frac{z + z^2 + z^3}{(z-2)(z-3)(z-4)(z-5)}$$

Solution : Given $f(z) = \frac{z + z^2 + z^3}{(z-2)(z-3)(z-4)(z-5)}$

Solving $(z-2)(z-3)(z-4)(z-5) = 0$, we get

$z = 2, 3, 4, 5$ as poles. All of these are simple pole.

Now residue at $z = 2$ is

$$\begin{aligned} \lim_{z \rightarrow 2} (z-2) \frac{z + z^2 + z^3}{(z-2)(z-3)(z-4)(z-5)} \\ &= \frac{2 + 4 + 8}{(-1)(-2)(-3)} \\ &= -\frac{14}{6} \\ &= -\frac{7}{3} \end{aligned}$$

residue at $z = 3$ is

$$\begin{aligned} \lim_{z \rightarrow 3} (z-3) \frac{z + z^2 + z^3}{(z-2)(z-3)(z-4)(z-5)} \\ &= \frac{3 + 9 + 27}{1 \cdot (-1)(-2)} \\ &= \frac{39}{2} \end{aligned}$$

residue at $z = 4$

$$\begin{aligned} \lim_{z \rightarrow 4} (z-4) \frac{z + z^2 + z^3}{(z-2)(z-3)(z-4)(z-5)} \\ &= \frac{4 + 16 + 64}{2 \cdot 1 \cdot (-1)} \\ &= -42 \end{aligned}$$

residue at $z = 5$

$$\begin{aligned} \lim_{z \rightarrow 5} (z-5) \frac{z + z^2 + z^3}{(z-2)(z-3)(z-4)(z-5)} \\ &= \frac{5 + 25 + 125}{3 \cdot 2 \cdot 1} \\ &= \frac{155}{6} \end{aligned}$$

4. Find the residues of the functions $f(z) = \frac{1}{z(z^2 + 1)(z - 2)^2}$

Solution : Given $f(z) = \frac{1}{z(z^2 + 1)(z - 2)^2}$

Solving $z(z^2 + 1)(z - 2)^2 = 0$, we find $z = 0, \pm i, 2$

$\therefore z = 0, z = \pm i$, are simple poles and $z = 2$ is a double pole.

Now residue at $z = 0$,

$$\begin{aligned}\lim_{z \rightarrow 0} z \cdot \frac{1}{z(z^2 + 1)(z - 2)^2} \\&= \frac{1}{1 \cdot (-2)^2} \\&= \frac{1}{4}\end{aligned}$$

residue at $z = i$,

$$\begin{aligned}\lim_{z \rightarrow i} (z - i) \frac{1}{z(z - i)(z + i)(z - 2)^2} \\&= \frac{1}{i \cdot 2i (i - 2)^2} \\&= \frac{1}{-2(i^2 - 4i + 4)}\end{aligned}$$

$$\begin{aligned}&= \frac{1}{-2(-1 - 4i + 4)} \\&= \frac{1}{-2(3 - 4i)} \\&= \frac{1}{8i - 6}\end{aligned}$$

Similarly, residue at $z = -i$ is

$$\frac{1}{-8i - 6} \quad [\text{Replacing } i \text{ by } -i \text{ in the residue at } z = i]$$

Residue at $z = 2$,

$$\begin{aligned}\lim_{z \rightarrow 2} \frac{d}{dz} \left\{ (z - 2)^2 \frac{1}{z(z^2 + 1)(z - 2)^2} \right\} \\&= \lim_{z \rightarrow 2} \frac{d}{dz} \left\{ \frac{1}{z^3 + z} \right\} \\&= \lim_{z \rightarrow 2} \frac{-(3z^2 + 1)}{(z^3 + z)^2} \\&= \frac{-(3 \cdot 4 + 1)}{(8 + 2)^2} \\&= \frac{-13}{100}\end{aligned}$$

Cauchy's Residue Theorem:

If $f(z)$ is analytic inside and on a simple closed curve C except at a finite number of points $a, b, c, \dots$ inside C , at which the residues are $a_{-1}, b_{-1}, c_{-1}, \dots$ respectively, then

$$\oint_C f(z) dz = 2\pi i(a_{-1} + b_{-1} + c_{-1} + \dots) \\ = 2\pi i \times (\text{sum of residues}).$$

1. Using Cauchy's residue theorem, evaluate the integral $\oint_C \frac{5z-2}{z(z-1)} dz$; where C is the circle $|z| = 2$.

Solution : Let $f(z) = \frac{5z-2}{z(z-1)}$

Solving $z(z-1) = 0$, we find $z = 0, 1$

$\therefore z = 0, 1$ are simple poles and lie inside C .

Residue at $z = 0$,

$$\lim_{z \rightarrow 0} z \cdot \frac{5z-2}{z(z-1)} \\ = \frac{0-2}{0-1} \\ = 2$$

Residue at $z = 1$,

$$\lim_{z \rightarrow 1} (z-1) \frac{5z-2}{z(z-1)} \\ = \frac{5-2}{1} \\ = 3$$

Therefore by Cauchy's residue theorem, we have

$$\oint_C \frac{5z-2}{z(z-1)} dz = 2\pi i (\text{sum of residues}) \\ = 2\pi i (2+3) \\ = 10\pi i.$$

2. Evaluate $\oint_C \frac{\sin z}{(z - \pi)^3} dz$; where C is the circle $|z| = 4$

Solution : Let $f(z) = \frac{\sin z}{(z - \pi)^3}$.

Solving $(z - \pi)^3 = 0$, we get $z = \pi$.

$\therefore z = \pi$ is a pole of order 3 and lie inside C.

Residue of $f(z)$ at $z = \pi$

$$\begin{aligned} & \frac{1}{(3-1)!} \lim_{z \rightarrow \pi} \frac{d}{dz^2} (z - \pi)^3 f(z) \\ &= \frac{1}{2!} \lim_{z \rightarrow \pi} \frac{d^2}{dz^2} (z - \pi)^3 \frac{\sin z}{(z - \pi)^3} \\ &= \frac{1}{2} \lim_{z \rightarrow \pi} \frac{d^2}{dz^2} (\sin z) \\ &= \frac{1}{2} \lim_{z \rightarrow \pi} (-\sin z) \\ &= 0 \end{aligned}$$

Therefore, by Cauchy's residue theorem, we get

$$\begin{aligned} \oint_C \frac{\sin z}{(z - \pi)^3} dz &= 2\pi i \text{ (sum of residues)} \\ &= 2\pi i \cdot 0 \\ &= 0. \end{aligned}$$

3. Evaluate $\oint_C \frac{z}{z^2 + 2z - 3} dz$; where C is the circle $|z| = 2$.

Solution : Let $f(z) = \frac{z}{z^2 + 2z - 3} = \frac{z}{(z - 1)(z + 3)}$.

Solving $(z - 1)(z + 3) = 0$, we get $z = 1, -3$

$\therefore z = 1$ and $z = -3$ are simple poles of $f(z)$ for which $z = 1$ lies inside C.

Residue of $f(z)$ at $z = 1$

$$\begin{aligned} & \lim_{z \rightarrow 1} (z - 1) f(z) \\ &= \lim_{z \rightarrow 1} (z - 1) \frac{z}{(z - 1)(z + 3)} \\ &= \lim_{z \rightarrow 1} \frac{z}{z + 3} \\ &= \frac{1}{4} \end{aligned}$$

Therefore, by Cauchy's residue theorem, we have

$$\begin{aligned} \oint_C \frac{z}{z^2 + 2z - 3} dz &= 2\pi i \text{ (sum of residues)} \\ &= 2\pi i \cdot \frac{1}{4} \\ &= \frac{\pi i}{2}. \end{aligned}$$

4. Evaluate $\oint_C \frac{\cos z}{z(z-1)^2} dz$ using Cauchy's Residue theorem,

where C is the circle $|z| = 2$.

Solution : Solving $z(z-1)^2 = 0$, we get $z = 0, 1$.

$\therefore z = 0$ is a simple pole and $z = 1$ is a double pole, both of them

lie inside C.

Residue of $f(z)$ at $z = 0$

$$\lim_{z \rightarrow 0} z f(z)$$

$$z \rightarrow 0$$

$$= \lim_{z \rightarrow 0} z \frac{\cos z}{z(z-1)^2}$$

$$= \lim_{z \rightarrow 0} \frac{\cos z}{(z-1)^2}$$

$$= \frac{1}{1}$$

$$= 1$$

Residue of $f(z)$ at $z = 1$

$$\frac{1}{(2-1)!} \lim_{z \rightarrow 1} \frac{d}{dz} [(z-1)^2 f(z)]$$

$$= \lim_{z \rightarrow 1} \frac{d}{dz} \left[(z-1)^2 \frac{\cos z}{z(z-1)^2} \right]$$

$$= \lim_{z \rightarrow 1} \frac{d}{dz} \left[\frac{\cos z}{z} \right]$$

$$= \lim_{z \rightarrow 1} \frac{z(-\sin z) - \cos z \cdot 1}{z^2}$$

$$= -\sin(1) - \cos(1)$$

$\therefore$ By Cauchy's residue theorem, we get

$$\oint_C \frac{\cos z}{z(z-1)^2} dz = 2\pi i (\text{sum of residues})$$

$$= 2\pi i (1 - \sin(1) - \cos(1)).$$

5. Evaluate $\oint_C \frac{z}{(z^2 + 1)(z^2 - 9)} dz$, using

Cauchy's residue theorem; where C is the circle $|z| = 2$.

Solution : Let $f(z) = \frac{z}{(z^2 + 1)(z^2 - 9)}$.

Solving $(z^2 + 1)(z^2 - 9) = 0$, we get $z = i, -i, 3, -3$.

$\therefore z = i, -i$ lie inside the circle C .

So, we have to calculate residues only at these two poles,

Residue of $f(z)$ at $z = i$

$$\begin{aligned} \lim_{z \rightarrow i} (z - i) f(z) &= \lim_{z \rightarrow i} (z - i) \frac{z}{(z + i)(z - i)(z + 3)(z - 3)} \\ &= \lim_{z \rightarrow i} \frac{z}{(z + i)(z^2 - 9)} \\ &= \frac{1}{(i + i)(-1 - 9)} \\ &= -\frac{1}{20} \end{aligned}$$

Residue of $f(z)$ at $z = -i$

$$\begin{aligned} \lim_{z \rightarrow -i} (z + i) f(z) &= \lim_{z \rightarrow -i} (z + i) \frac{z}{(z + i)(z - i)(z + 3)(z - 3)} \\ &= \lim_{z \rightarrow -i} \frac{z}{(z - i)(z^2 - 9)} \\ &= \lim_{z \rightarrow -i} \frac{-i}{(-2i)(-10)} \\ &= -\frac{1}{20} \end{aligned}$$

$\therefore$ By Cauchy's residue theorem, we have

$$\oint_C \frac{z}{(z^2 + 1)(z^2 - 9)} dz = 2\pi i \left(-\frac{1}{20} - \frac{1}{20} \right) = -\frac{\pi i}{5}$$

Problem-1

Show that

$$\oint_C \frac{e^z}{(z^2 + \pi^2)^2} dz = \frac{1}{\pi i}$$

where C is the circle $|z| = 4$.

Solution:

Let

$$f(z) = \frac{e^z}{(z + \pi i)^2(z - \pi i)^2}$$

$z = \pi i$ and $z = -\pi i$ are two poles of $f(z)$, each of them being double poles. Since $|\pi i| = |-\pi i| = \pi < 4$, these poles are inside C .

Residue at $z = \pi i$:

$$\begin{aligned} \text{Residue} &= \lim_{z \rightarrow \pi i} \frac{1}{1!} \frac{d}{dz} \left\{ (z - \pi i)^2 \frac{e^z}{(z + \pi i)^2(z - \pi i)^2} \right\} \\ &= \lim_{z \rightarrow \pi i} \frac{e^z(z + \pi i)^2(2i\pi - 2)}{(z + \pi i)^4} \\ &= \frac{e^{\pi i}(2i\pi - 2)}{(2i\pi)^3} \\ &= \frac{-(\pi + i)}{4\pi^3} e^{\pi i} \end{aligned}$$

Similarly, residue at $z = -\pi i$:

$$\frac{-(\pi - i)}{4\pi^3} e^{-\pi i}$$

Applying Cauchy's Residue Theorem:

$$\begin{aligned} \oint_C \frac{e^z}{(z^2 + \pi^2)^2} dz &= 2\pi i \times (\text{sum of residues}) \\ &= 2\pi i \left[\frac{-(\pi + i)}{4\pi^3} e^{\pi i} + \frac{-(\pi - i)}{4\pi^3} e^{-\pi i} \right] \\ &= \frac{i}{2\pi^2} [-\pi(e^{\pi i} + e^{-\pi i}) - i(e^{\pi i} - e^{-\pi i})] \\ &= \frac{i}{2\pi^2} [-2\pi \cos \pi - i \cdot 2i \sin \pi] \\ &= \frac{i}{2\pi^2} [-2\pi(-1) - 2i^2 \cdot 0] \\ &= \frac{i}{\pi} \end{aligned}$$

(d) Evaluate the following integrals, using Cauchy's residue theorem.

1. $\oint_C \frac{z^3}{z+1} dz$; C is $|z| = 2$
2. $\oint_C \frac{dz}{z-2}$; C is $|z-2| = 4$
3. $\oint_C \frac{dz}{z(z^2+4)}$; C is $|z| = 1$
4. $\oint_C \frac{dz}{z^2(z+4)}$; C is $|z| = 5$
5. $\oint_C \frac{z dz}{(z+1)(9-z^2)}$; C is $|z| = 2$.
6. $\oint_C \frac{2z+1}{z^2+z-6} dz$; C is $|z| = 4$
7. $\oint_C \frac{2z+1}{(z-2)(z+3)(z+1)} dz$; C is $|z| = \frac{5}{2}$.

Problem 3: Square Contour Integrals

Let C denote the square in the complex plane bounded by the lines $x = \pm 2$ and $y = \pm 2$, traversed in the positive (counter-clockwise) sense. Evaluate the following integrals:

(i)

$$\oint_C \frac{dz}{z^2 + 9}$$

(ii)

$$\oint_C \frac{dz}{z(z^2 + 9)}$$

(iii)

$$\oint_C \frac{dz}{(z^2 + 1)(z^2 + 9)}$$